MemOS (Enterprise Memory Operating System)

Vision

The AI brain that remembers everything an organization knows.

Overall Architecture

React Frontend -> FastAPI Backend -> LangGraph Supervisor -> Search/Graph/Timeline/Decision/Review Agents -> Enterprise Memory Layer (Neo4j, Qdrant, PostgreSQL, Object Storage) -> Document Pipeline -> GitHub, Slack, Jira, Docs, Email, Teams.

Phase 1 – Enterprise Memory Core

Upload PDFs, DOCX, Markdown and TXT. Parse, chunk, embed into Qdrant, extract entities into Neo4j. Features: semantic search, chat with docs, graph visualization, citations.

Phase 2 – Knowledge Graph

Build relationships between people, projects, APIs, meetings, decisions and technologies. Add graph explorer and dependency search.

Phase 3 – GitHub Intelligence

Ingest repositories, commits, PRs, issues and releases. Enable repository reasoning and expert discovery.

Phase 4 – Decision Memory

Extract decisions, alternatives, reasons, owners and outcomes from meetings and documents so AI remembers WHY, not just WHAT.

Phase 5 – Timeline Engine

Automatically reconstruct incident timelines using commits, deployments, Slack, customer complaints and rollbacks.

Phase 6 – Multi-Agent Brain

Supervisor coordinates Search, Graph, Timeline, Decision, Code and Reviewer agents.

Phase 7 – Continuous Learning

Automatically ingest new emails, Slack messages, meetings, commits and documents.

Phase 8 – Integrations

Connect GitHub, Jira, Slack, Notion, Confluence, Google Drive, Outlook, Teams, Linear and ClickUp.

Phase 9 – Enterprise Intelligence

Correlate engineering, customer and business data to discover insights automatically.

Phase 10 – Enterprise Copilot

Answer complex organizational questions by reasoning over documents, graph, timelines and decisions.

Phase 11 – Autonomous Organization

Detect events (e.g., vulnerabilities), create tasks, notify owners and track remediation automatically.

Stretch Goals

Knowledge evolution, conflict detection, expertise graph, counterfactual reasoning, decision replay and enterprise simulation.